

Modified DeCoR under the Asymmetric Confounding: Sparse to Predictor, Dense to Response

Seminar Advances in Statistical Inference - Prof. Drton

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Abstract

This report modifies the DeCoR algorithm to address an asymmetric confounding scenario mentioned in the future work section of [1]. Specifically, a setting is investigated where a hidden confounder acts densely on the response variable but sparsely on the predictor in the frequency domain. It will be empirically demonstrated that the standard DeCoR fails in this setting. Afterwards, we show the frequency-domain orthogonality condition between the predictor and the confounder (mentioned in [1], $(T(X))_k^\top (T(U))_k \xrightarrow{P} 0 \quad \forall k \in G^c$). Then we utilize this orthogonality condition, by modifying the Torrent algorithm's [2] initialization of inliers' set and its iterative Hard-Thresholding loop and we see that the relationship β can be more accurately recovered than standard OLS as the sample size n increases. We empirically validate the modified algorithm using AR(1)-processes following the simulation designs of the original work [1].

1 Introduction

The original work [1] studied the problem of estimating the causal relationship β between predictor and response in the following structural setup:

$$Y = X\beta + U + \epsilon$$

Here, X, Y, U , and ϵ correspond to the predictor, response, hidden confounder, and i.i.d. centered Gaussians respectively. They assumed the setting described in the left graph of Figure 1, in which the confounder can act dense on the predictor and must act sparse on the response. This allowed them to consider this estimation problem as an outlier problem in frequency domain, as the data points corresponding to frequencies in the confounded set G would have higher residuals due to the present confounder in the composition of the response Y :

$$T(Y)_k \stackrel{\text{a.s.}}{=} \begin{cases} (T(X)\beta)_k + T(U)_k + T(\epsilon)_k & \text{if } k \in G, \\ (T(X)\beta)_k + T(\epsilon)_k & \text{if } k \notin G. \end{cases}$$

where T corresponds to the frequency domain transformation function. To exploit this tendency of the outlier points to have high residuals, the DecoR-Torrent algorithm [1] utilized the Hard-Thresholding step in the iterative loop, where the set of inliers is updated by throwing out the points with the highest residuals.

In the new setting described in the right graph of Figure 1, confounder can be present inside the response in all frequencies. This prevents us from using the same algorithm to estimate the relationship β accurately, which is visualized in Figure 2. Here we simulate the following structural setup that corresponds to the right graph of Figure 1:

$$\begin{aligned} Y &= X^\top \beta + U + \epsilon_Y, \\ X &= Q + U_{\text{sparse}} + \epsilon_X \end{aligned}$$

where $X, Y, U \in \mathbb{R}^n$ and Q and U correspond to two independent stationary AR(1)-processes, similar to Sec. 6.1 of [1]. U_{sparse} corresponds to the U projected to a small set of outlier frequencies belonging to a confounded set G . In this setting, we see in Figure 2 that DecoR-Torrent fails to filter out outlier frequencies and rather randomly picks the frequencies to filter out. Moreover, it performs even worse than standard OLS, filtering out intact inliers and estimating the relationship β using less data points as OLS. This limitation motivates us to modify the algorithm to address this new scenario.

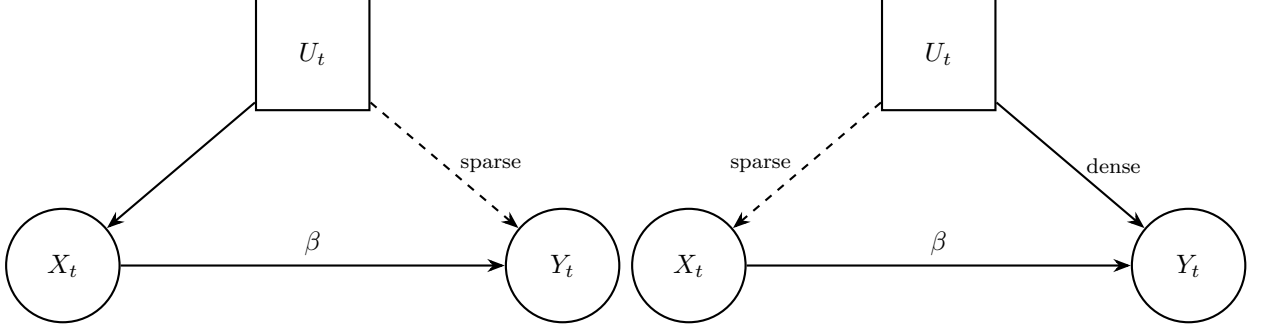


Figure 1: Left: Scenario discussed in original work [1].
Right: Changed setup, where the confounder U_t acts sparsely on X_t but densely on Y_t .

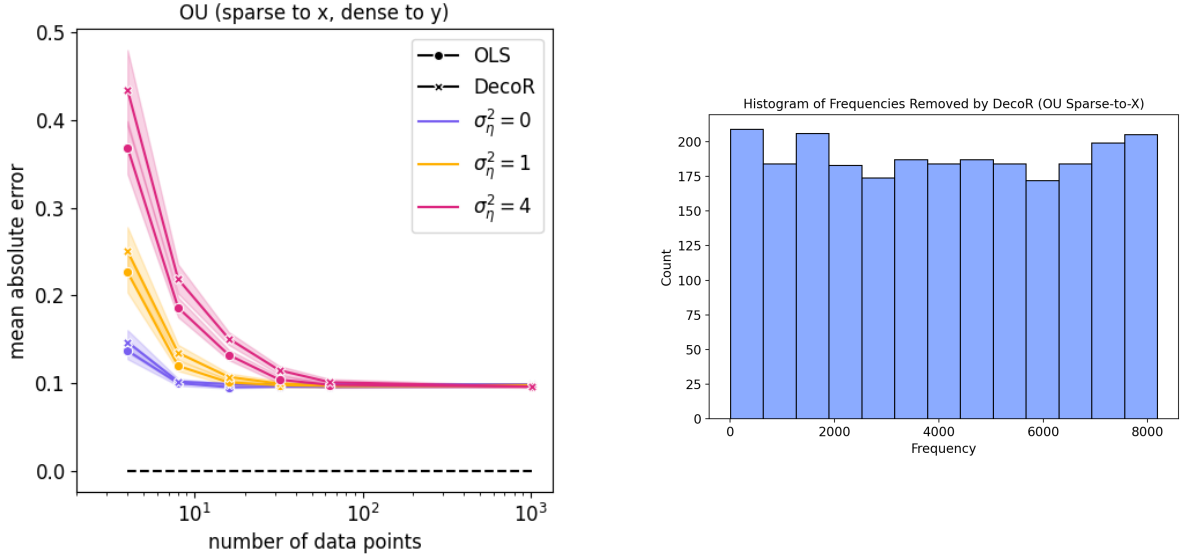


Figure 2: Failing of DeCoR in the new setting.
Left: Mean absolute estimation errors, $\frac{1}{m} \sum_i^m |\hat{\beta}_i - \beta|$, of m -times repeated simulations for OLS and DecoR. Right: Removed outlier frequencies by DecoR.

2 Theoretical Framework

For simplicity, we consider one-dimensional predictor X , confounder U , and response Y . We assume the following structural equations:

$$Y = \beta \cdot X + U + \epsilon_Y \quad (1)$$

$$X = \alpha \cdot U_{sparse} + \epsilon_X \quad (2)$$

where ϵ_X, ϵ_Y are Gaussian noise vector with zero mean and covariance $\sigma^2 \cdot I_n$ and the relationship β and α are scalars. X, Y , and U are n -dimensional vectors in $\mathbb{R}^n$. Following the setup of [1], $U \in \mathbb{R}^n$ is generated using a (covariance) stationary AR(1)-process with the following equation:

$$U_t = \varphi U_{t-1} + \epsilon_t$$

$$\forall t \in [n], \quad \varphi = \frac{0.8}{n}, \quad \epsilon_t \sim N(0, \sigma_\epsilon^2 = 1/n)$$

Now we consider the orthogonality claim made in the future work section of [1], which will be the basis of our algorithmic modifications.

Claim

$$(T(X))_k^\top (T(U))_k \xrightarrow{P} 0 \quad \forall k \in G^c \quad (3)$$

where $T(X) = \frac{1}{n} \Phi^\top(X)$ is the frequency domain transformation function with Φ being an orthogonal matrix and G^c being the complement of the confounded frequency set G .

Proof: For any clean (unconfounded) frequency index $k \notin \text{supp}(T(U_{\text{sparse}}))$ by evaluating Eq. 2:

$$\begin{aligned} (T(X))_k^\top (T(U))_k &= (\alpha \cdot T(U_{\text{sparse}}) + T(\epsilon_X))_k^\top (T(U))_k \\ &= \alpha \cdot T(U_{\text{sparse}})_k^\top T(U)_k + T(\epsilon_X)_k^\top T(U)_k \\ &= \alpha \cdot 0 \cdot T(U)_k + T(\epsilon_X)_k^\top T(U)_k \\ &= T(\epsilon_X)_k^\top T(U)_k \end{aligned}$$

The second last equation follows directly from the sparsity of the confounder on the inlier set. Now, we write out the terms after transformations explicitly:

$$T(\epsilon_X)_k = \frac{1}{n} \Phi_k^\top \epsilon_X, \quad (4)$$

$$T(U)_k = \frac{1}{n} \Phi_k^\top U, \quad (5)$$

where Φ_k denotes the k -th column of the orthogonal basis matrix Φ . Plugging this in, their inner product expands as:

$$\begin{aligned} T(\epsilon_X)_k^\top T(U)_k &= \left(\frac{1}{n} \Phi_k^\top \epsilon_X \right)^\top \left(\frac{1}{n} \Phi_k^\top U \right) \\ &= \frac{1}{n^2} \left(\epsilon_X^\top \Phi_k \right) \left(\Phi_k^\top U \right) \end{aligned}$$

Looking at the individual terms separately:

- We know $\epsilon_X \sim N(\mathbf{0}, \sigma^2 I_n)$. Then $\Phi_k^\top \epsilon_X \sim N(\Phi_k^\top \mathbf{0}, \Phi_k^\top (\sigma^2 I_n) \Phi_k) = N(0, \sigma^2)$, using Φ_k being normalized. Therefore, $\Phi_k^\top \epsilon_X = O_p(1)$
- – Next, we notice that our AR(1)-process from Eq. 5 is stationary. This can be either directly argued by noticing that the magnitude of the one-lag coefficient is strictly upper-bounded by 1, i.e $|\varphi| = \frac{0.8}{n} < 1$ or checking that the complex root of our characteristic polynomial $1 - \varphi z$ is outside the unit-circle. Indeed, our scalar root $z = \frac{n}{0.8}$ has an absolute value greater than 1 as $n \geq 1$.

- Using stationarity, i.e. $\mathbb{E}(U_t) = \mu \forall t \in [n]$ and zero mean noise assumptions, we find $\mathbb{E}(U_t) = 0 \forall t \in [n]$ and therefore, $\mathbb{E}(U) = \mathbf{0} \in \mathbb{R}^n$:

$$\begin{aligned}\mu &= \mathbb{E}(U_t) = \mathbb{E}(\varphi \cdot U_{t-1} + \epsilon_t) \\ &= \frac{0.8}{n} \mu + \mathbb{E}(\epsilon_t) \\ &= \frac{0.8}{n} \mu \quad \forall n \in \mathbb{N} \\ &\iff \mu = 0\end{aligned}$$

It follows naturally that $\mathbb{E}(\Phi_k^\top U) = 0$.

- Now, we consider the covariance matrix $\text{Var}(U)$. Following [3], we have $\text{Var}(U_t) = \frac{\sigma_\epsilon^2}{1-\varphi^2} > 0$ on the diagonal. On the off-diagonal, we have $\gamma_k := \text{Cov}(U_t, U_{t-k}) = \frac{\varphi^k}{1-\varphi^2} \sigma_\epsilon^2 > 0$ with $\gamma_k = \gamma_{-k}$ denoting the autocovariance at lag k . Here, we further note that $\gamma_k = \gamma_0 \cdot \varphi^{|k|}$ with $\gamma_0 = \text{Var}(U_t) \quad \forall t \in [n]$.
- Let $R_i := \sum_{j \neq i}^n |\text{Cov}(U)_{ij}|$ be the sum of the absolute values of the non-diagonal entries in the i -th row. Then:

$$\begin{aligned}R_i &= \sum_{j \neq i} |\text{Cov}(U_i, U_j)| < \sum_{j=1}^n |\text{Cov}(U_i, U_j)| = \sum_{j=1}^n |\gamma_{i-j}| = \sum_{k=i-n}^{i-1} |\gamma_k| \\ &< \sum_{k=-\infty}^{\infty} |\gamma_k| = \sum_{k=-\infty}^{\infty} \gamma_0 \varphi^{|k|} = \sum_{k=-\infty}^{\infty} \gamma_0 \varphi^{|k|} \\ &= \gamma_0 \left(\sum_{k=-\infty}^{-1} \varphi^{-k} + \varphi^0 + \sum_{k=1}^{\infty} \varphi^k \right) \\ &= \gamma_0 \left(1 + 2 \sum_{k=1}^{\infty} \varphi^k \right) = \gamma_0 \left(1 + 2 \left(\frac{\varphi}{1-\varphi} \right) \right) = \gamma_0 \left(\frac{1+\varphi}{1-\varphi} \right)\end{aligned}$$

Substituting $\gamma_0 = \frac{\sigma_\epsilon^2}{1-\varphi^2} = \frac{\sigma_\epsilon^2}{(1-\varphi)(1+\varphi)}$ and plugging in the parameters of the used AR(1)-process, the expression simplifies to:

$$\begin{aligned}R_i &< \sum_{k=-\infty}^{\infty} |\gamma_k| = \frac{\sigma_\epsilon^2}{(1-\varphi)(1+\varphi)} \left(\frac{1+\varphi}{1-\varphi} \right) = \frac{\sigma_\epsilon^2}{(1-\varphi)^2} \\ &= \frac{1/n}{(1 - \frac{0.8}{n})^2} = \frac{n}{(n-0.8)^2} = O(n^{-1}) \quad \forall i \in [n]\end{aligned}$$

Then, by Gerschgorin Circle Theorem, the maximum eigenvalue $\lambda_{\max}(\Sigma_U)$ of $\text{Var}(U)$ is finite and decays at a rate of $O(n^{-1})$.

- Therefore, $\text{Var}(\Phi_k^\top U) = \Phi_k^\top \text{Var}(U) \Phi_k \leq \lambda_{\max}(\Sigma_U) \cdot \|\Phi_k\|_2^2 = O(n^{-1}) =: \frac{C}{n}$. Now, we use the Chebyshev inequality to bound $\Phi_k^\top U$:

$$\mathbb{P}(|\Phi_k^\top U| \geq Mn^{-1/2}) \leq \frac{C/n}{(Mn^{-1/2})^2} = \frac{C}{M^2}, \quad \forall n \in \mathbb{N}$$

Therefore: $\Phi_k^\top U = O_p(n^{-1/2})$

- Plugging in the decaying rate of the prefactor $\frac{1}{n^2} = O(n^{-2})$ yields:

$$T(\epsilon_X)_k^\top T(U)_k = \frac{1}{n^2} \left(\epsilon_X^\top \Phi_k \right) \left(\Phi_k^\top U \right) = O(n^{-2}) \cdot O_p(1) \cdot O_p(n^{-1/2}) = O_p(n^{-2.5})$$

i.e. $T(\epsilon_X)_k^\top T(U)_k \xrightarrow{P} 0$.

3 Methodology

Now, we will use the property of Eq. 3 to modify the Torrent algorithm and to filter the inlier set in the changed scenario. For notational ease, we denote the frequency-transformed data as follows:

$$T(X) =: X^\phi, \quad T(Y) =: Y^\phi, \quad T(U) =: U^\phi$$

With the superscript ϕ signaling the basis of the transformed frequency domain.

Algorithm 1 Original Torrent Algorithm [1]

Require: $X^\phi \in \mathbb{R}^n, Y^\phi \in \mathbb{R}^n, a := (n - c) \leq |G^c|$

- 1: **Initialize:** $S_0 \leftarrow \{1, \dots, n\}, \text{currErr} \leftarrow \infty, t \leftarrow 0$ $\triangleright$ Initialize inlier set with all data points
- 2: **repeat**
- 3: $\hat{\beta} \leftarrow \hat{\beta}_{\text{OLS}}(X_{S_t}^\phi, Y_{S_t}^\phi)$ $\triangleright$ Fit OLS on current S_t
- 4: $e \leftarrow |Y^\phi - X^\phi \hat{\beta}|$ $\triangleright$ Calc residuals for ALL data points
- 5: $t \leftarrow t + 1$
- 6: $S_t \leftarrow \text{HT}(e, a)$ $\triangleright$ Keep a points with smallest residuals
- 7: $\text{lastErr} \leftarrow \text{currErr}$
- 8: $\text{currErr} \leftarrow \|Y_{S_t}^\phi - X_{S_t}^\phi \hat{\beta}\|_2$ $\triangleright$ Check whether we improved or not
- 9: **until** $\text{currErr} \geq \text{lastErr}$ $\triangleright$ Keep going if we improved
- 10: **return** $\hat{\beta}_{\text{Tor}}^t$

Algorithm 2 Modified Torrent Algorithm

Require: $X^\phi \in \mathbb{R}^n, Y^\phi \in \mathbb{R}^n, a := (n - c) \leq |G^c|$

- 1: $v \leftarrow |X^\phi \odot Y^\phi|$
- 2: $S_0 \leftarrow \text{HT}(v, a), t \leftarrow 0$ $\triangleright$ Initialize with a points with smallest values in v
- 3: **repeat**
- 4: $\hat{\beta} \leftarrow \hat{\beta}_{\text{OLS}}(X_{S_t}^\phi, Y_{S_t}^\phi)$ $\triangleright$ Fit OLS on active set
- 5: $e \leftarrow Y^\phi - X^\phi \hat{\beta}$ $\triangleright$ Calc residuals for ALL data points
- 6: $s \leftarrow |X^\phi \odot e|$ $\triangleright$ Evaluate point-wise orthogonality
- 7: $t \leftarrow t + 1$
- 8: $S_t \leftarrow \text{HT}(s, a)$ $\triangleright$ Keep a points with smallest residuals
- 9: **until** $S_t = S_{t-1}$ $\triangleright$ Stop if our set did not change
- 10: **return** $\hat{\beta}$

The goal of the original Torrent algorithm (see Algo.1) is to retrieve a clean set of inliers (without outliers) to fit a OLS line on them. Here, the key steps consist of its initialization of this inlier set, i.e. the set of data points we believe to be inliers, and the Hard-Thresholding step. The initialization is done by simply adding all data points into the inlier set. Then, the Hard-Thresholding step looks at the residuals, which are computed using the current estimate $\hat{\beta}$, and keeps only a many points with the smallest residuals.

To address our specific scenario of confounding, we change these two key steps of the original algorithm (see Algo. 2):

1. Initialization:

In the Modified Torrent Algorithm 2, we initialize the first inlier set S_0 with data points yielding smallest values of $|X^\phi \odot Y^\phi|$. To follow the reasoning of this strategy, we look at

$|(X^\phi \odot Y^\phi)_k|$, distinguishing the case when $k \in G^c$ and not:

$$\begin{aligned} X_k^\phi \cdot Y_k^\phi &= X_k^\phi (X_k^\phi \cdot \beta + U_k^\phi + (\epsilon_Y^\phi)_k) \\ &= (X_k^\phi)^2 \cdot \beta + X_k^\phi U_k^\phi + X_k^\phi (\epsilon_Y^\phi)_k \\ \implies \mathbb{E}(X_k^\phi \cdot Y_k^\phi) &= \beta \cdot \mathbb{E}((X_k^\phi)^2) + \mathbb{E}(X_k^\phi U_k^\phi) + \mathbb{E}(X_k^\phi (\epsilon_Y^\phi)_k) \end{aligned}$$

Recall: $X = \alpha U_{\text{sparse}} + \epsilon_X$

- if $k \in \text{inliers } G^c$:

$$\begin{aligned} k \in G^c &\implies X_k^\phi = (\epsilon_X^\phi)_k \\ \mathbb{E}(X_k^\phi \cdot Y_k^\phi) &= \beta \cdot \mathbb{E}((X_k^\phi)^2) + \mathbb{E}(X_k^\phi U_k^\phi) + \mathbb{E}(X_k^\phi (\epsilon_Y^\phi)_k) \\ &= \beta \cdot \mathbb{E}((\epsilon_X^\phi)_k^2) + \underbrace{\mathbb{E}((\epsilon_X^\phi)_k U_k^\phi)}_{=0} + \underbrace{\mathbb{E}((\epsilon_X^\phi)_k (\epsilon_Y^\phi)_k)}_{=0} \\ &= \beta \cdot \sigma^2 \end{aligned}$$

- if $k \in \text{outliers } G$:

$$\begin{aligned} k \in G &\implies X_k^\phi = \alpha (U_{\text{sparse}}^\phi)_k + (\epsilon_X^\phi)_k \\ \mathbb{E}(X_k^\phi \cdot Y_k^\phi) &= \beta \cdot \mathbb{E}((X_k^\phi)^2) + \mathbb{E}(X_k^\phi U_k^\phi) + \mathbb{E}(X_k^\phi (\epsilon_Y^\phi)_k) \\ &= \beta \cdot \mathbb{E}((\alpha (U_{\text{sparse}}^\phi)_k + (\epsilon_X^\phi)_k)^2) + \mathbb{E}((\alpha (U_{\text{sparse}}^\phi)_k + (\epsilon_X^\phi)_k) U_k^\phi) \\ &\quad + \underbrace{\mathbb{E}((\alpha (U_{\text{sparse}}^\phi)_k + (\epsilon_X^\phi)_k) (\epsilon_Y^\phi)_k)}_{=0} \\ &= \alpha(\alpha\beta + 1) \mathbb{E}((U_{\text{sparse}}^\phi)_k^2) + \beta \cdot \underbrace{\mathbb{E}((\epsilon_X^\phi)_k^2)}_{=\sigma^2} \end{aligned}$$

Therefore, this fact that $\mathbb{E}(|(X^\phi \odot Y^\phi)_i|) > \mathbb{E}(|(X^\phi \odot Y^\phi)_j|)$ with $i \in G^c$ and $j \in G$ motivates us to initialize the inlier set S_0 to contain those data points with smallest values of $\mathbb{E}(|(X^\phi \odot Y^\phi)_k|)$.

2. Iterative Hard-Thresholding:

We saw in Eq. 3 that $X_k^\phi \cdot U_k^\phi \rightarrow 0$ (in the new notation). This motivates us to use this property as a new Hard-Thresholding scheme to filter out the outliers. However, one obstacle here is that U is a hidden confounder we do not have access to. Therefore, we try to approximate U with the residuals $e := Y^\phi - X^\phi \hat{\beta}$. For this, we use the following alternating optimization scheme. Recall that the computed residuals decompose as $e = X^\phi(\beta - \hat{\beta}) + U^\phi + \epsilon_Y^\phi$.

- **Optimizing S_t :** Given the current estimate $\hat{\beta}$, we approximate U^ϕ with e , assuming $\hat{\beta} \approx \beta$. Using e as approximation of U^ϕ , we optimize the inlier set S_t using Hard-Thresholding.
- **Optimizing $\hat{\beta}$:** Given the current inlier set S_t , we optimize $\hat{\beta}$ using the closed-form solution of OLS.

Remark: In the modified Torrent, we see the problem as an optimization over a discrete subset space to find the optimal inlier set. Matching this objective, we terminate when the active set converged to a fixed point, i.e. $S_t = S_{t-1}$. Indeed, the original work [1] also implements their DecoR-Torrent using this stopping criterion in their provided Python code. Moreover, by the modified initialization from before, our starting inlier set S_0 should have filtered out at least the extreme outliers, helping this approximation of U .

4 Results

Here, we simulate the processes following the setting in Sec. 2. Figure 3 compares the mean absolute error, i.e. $\frac{1}{m} \sum_{i=1}^m |\hat{\beta}_i - \beta|$ with $m = 10$ being the number of repeated simulations, of OLS against both the original DecoR-Torrent and the modified DecoR-Torrent with growing sample sizes n . In the left panel of Figure 3, we see that standard OLS expectedly exhibits a persistent bias of ca. 0.15, being incapable of separating the hidden confounder. Here, the original DecoR-Torrent also fails to resolve the confounding and plateaus at a similar error level to OLS, due to the same reason mentioned in the Introduction. In the right panel, we observe an improved behavior from the modified DecoR-Torrent algorithm for large sample sizes. For small sample sizes ($n \lesssim 100$), the modified algorithm shows very high variance and performs worse than OLS. This instability is rather expected due to the following reasons. modified DecoR-Torrent uses less samples to estimate β than OLS, which could inflate the variance of the estimator, especially when n is small. Also, at small sample sizes, the scaling factor of $n^{-2.5}$ in our asymptotic orthogonality in Eq. 3 could be not yet small enough to suppress and make itself well distinguishable from noise. However, as n grows over 10^2 , the modified DecoR-Torrent performs better than OLS, supported by our asymptotic orthogonality. We can conclude, that it gradually isolated the clean frequencies to draw a more accurate estimate of β .

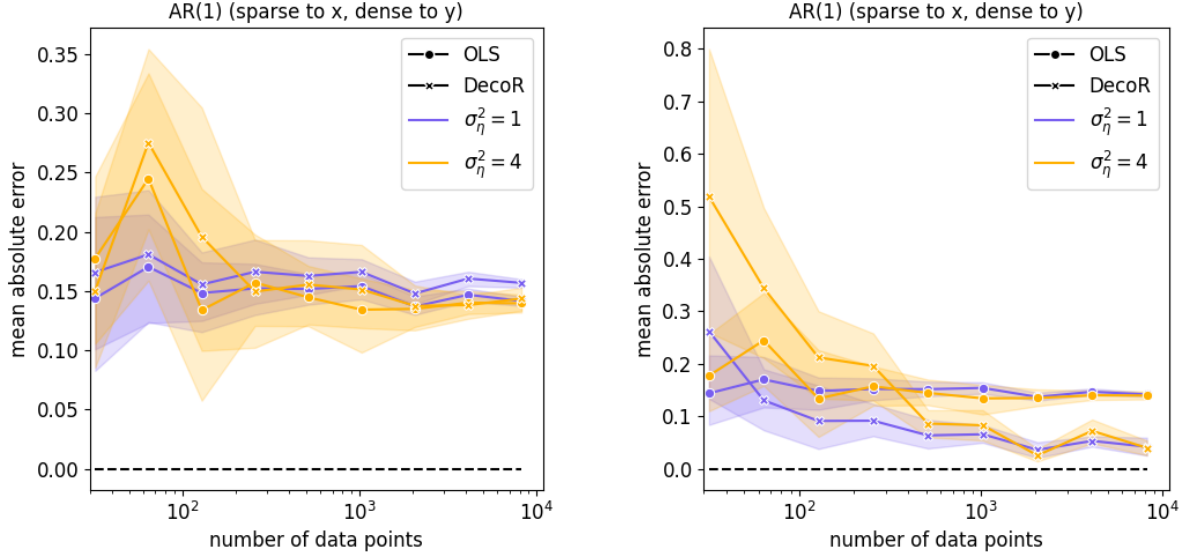


Figure 3: Comparison of original and modified DecoR-Torrent in the new scenario.

Left: OLS vs original DecoR-Torrent, Right: OLS vs modified DecoR-Torrent.

The simulated AR(1)-process follows the parametrization of Sec. 2.

5 Conclusion

This report addressed a limitation of the original DeCoR algorithm under an asymmetric confounding scenario, where a hidden confounder acts densely on the response but sparsely on the predictor. Empirical evaluations demonstrated that standard DeCoR fails under these conditions, performing worse than Ordinary Least Squares (OLS) by incorrectly filtering intact inliers. To overcome this, we showed an asymptotic frequency-domain orthogonality condition between the predictor and the confounder $((T(X))_k^\top (T(U))_k \xrightarrow{P} 0)$. This property has been utilized to update the initialization of the algorithm and to employ a modified alternating optimization scheme. This scheme uses model residuals to approximate the confounder and performs

iterative hard-thresholding based on point-wise orthogonality. Simulations under AR(1) processes successfully validated our approach. While exhibiting expected high-variance instability at small sample sizes ($n \leq 100$), the modified algorithm successfully isolates clean frequencies and performs better than OLS.

6 Code Availability

Code to reproduce simulations is available at: https://github.com/rlawnsiiiiii/robust_deconfounding

References

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